Graph of a Two-Variable Function

Question: How can we plot a surface?

- For $s = g(t)$, we pick points $t_1, \ldots, t_n$ and connect points with a smooth curve.
- For $z = f(x, y)$, this method fails. Instead:
  - For each $z = z_0$, draw the **level curve** $z_0 = f(x, y)$.
  - This is the **intersection** of the surface $z = f(x, y)$ and the horizontal plane $z = z_0$:

$$L_{z_0} := \{(x, y) \mid f(x, y) = z_0\}.$$

Slicing at $z = z_0$ and projecting contours to the xy-plane